Diagnosing Prototype Estimation in Few-Shot Class-Incremental Learning: Training-Free Geometry Selection and Unlabeled-Query Prototype Rectification

Md Tanvir Hossain Saikat[1] Md Akib Haider[1*] Md. Tahmid Islam Tomal[2,*]
Md Hasanul Kabir[1]
[1]Islamic University of Technology, Bangladesh
[2]Bangladesh University of Engineering and Technology, Bangladesh
{tanvirhossain3, akibhaider, hasanul}@iut-dhaka.edu
tahmid.tomal@icloud.com

Abstract

Freezing a visual encoder prevents representation drift, but a novel class must still be represented using only a few labeled examples. We study how much this uncertainty in class means contributes to the accuracy gap between a five-shot cosine classifier and a complete-data linear probe. On identical frozen features, we first replace novel means with means from the complete training split, keeping base prototypes and cosine scoring fixed. We then fit the linear probe. Across sixteen strong encoder–dataset pairs, mean replacement accounts for more than half of this ordered gap; a weaker base-only encoder does not show the same pattern. Motivated by this finding, ShrinkSel selects prototype radius and covariance shrinkage using base-class episodes, with regularized shared-covariance LDA as a fallback. In the sixteen-pair evaluation, it improves over LDA on fifteen pairs by 0.04–1.52 *percentage points and selects LDA on Aircraft. Query Prototype Rectification (*QPR*) additionally uses an unlabeled cohort to update novel prototype directions while preserving their radii and the selected metric. On the three standard benchmarks with CLIP ViT-B/16, it adds* 0.37–0.67 *points over cosine NCM and* 1.21–2.03 *points over ShrinkSel in session-average accuracy. Gains remain positive when each query is excluded from its own update. However, novel-class gains can come at the expense of base accuracy, and the benefit diminishes with more labeled supports or very small query cohorts.*

1. Introduction

Few-shot class-incremental learning (FSCIL) adds new classes from a small labeled support set while retaining the ability to recognize previously learned classes [27, 39]. Many methods modify the representation or classifier to balance these objectives [26, 37, 38, 40]. Though freezing the encoder removes representation drift, but accuracy can still decline as new classes enter the candidate set.

A prototype classifier illustrates the remaining difficulty. Base-class means are estimated from many examples, whereas each novel-class mean may use only five. An inaccurate novel mean can move a decision boundary even when the underlying features are useful. We ask how much accuracy can be recovered by improving these means before changing the classifier.

We examine this question through two comparisons on identical frozen features. First, we replace five-shot novel means with means computed from the complete training split, keeping base prototypes and cosine scoring fixed. Second, we fit a linear probe to the same complete training features. Mean replacement accounts for more than half of this ordered gap across sixteen strong CLIP, MAE-ft, and supervised ResNet settings. Shot-count and classwise diagnostics support a finite-sample explanation. A weaker base-only ResNet does not show the same majority pattern, which limits the finding to the representations and comparison studied here.

The diagnosis motivates two methods with different data requirements (Fig. 1). The support-only method, ShrinkSel, selects a prototype radius and covariance metric using base-class episodes. Cosine NCM and regularized shared-covariance LDA are classification endpoints, and a held-out check can return the selector to LDA. When an unlabeled query cohort is available, QPR also updates novel prototype directions. It preserves base prototypes, novel radii, and the selected metric. We evaluate the update with both ShrinkSel and cosine NCM to separate its contribution from the choice of support-only head. Neither prediction method uses text, complete novel-class labels, or known query-class proportions. Complete novel training splits are used only in the

*Equal contribution

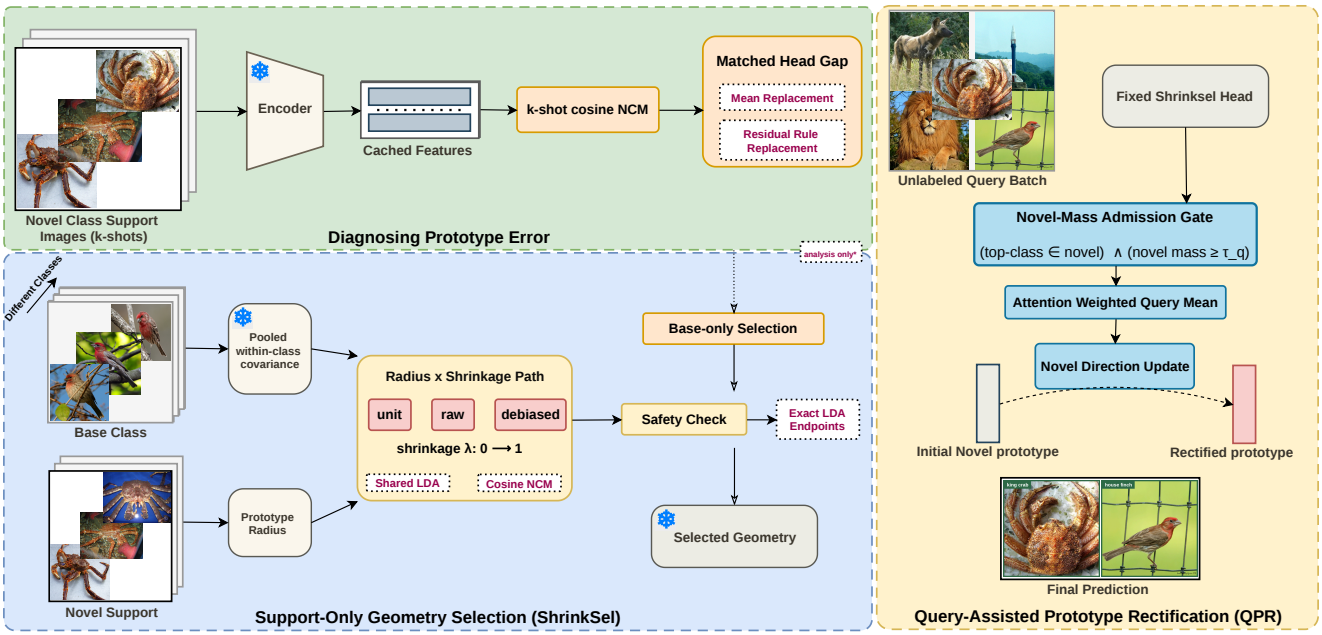

Figure 1. Overview of the analysis and the two prediction methods, all using frozen image features. The retrospective analysis replaces five-shot novel means with means from the complete training split, then fits a linear probe. ShrinkSel uses base classes to select a support-only classifier. QPR adds an unlabeled query cohort to refine novel prototype directions. Only the analysis uses complete novel training labels; neither prediction method uses query labels or known query-class proportions.

retrospective analysis.

Our contributions are:

- A controlled comparison of novel-mean replacement and classifier replacement on frozen features, with uncertainty estimates and supporting shot-count and classwise diagnostics.
- ShrinkSel, a support-only procedure that selects prototype radius and covariance shrinkage using base-class episodes, with an LDA fallback.
- QPR, a query-assisted direction update compared across two support-only heads and evaluated under query self-exclusion, varying cohort sizes, and base–novel accuracy trade-offs.

2. Related Work

FSCIL with fixed or adapted representations. FSCIL methods commonly modify the representation or classifier to preserve old classes and accommodate scarce new ones: CEC [38] evolves classifiers, FACT [40] reserves forward-compatible space, and NC-FSCIL [37], OrCo [1], and ASP [15] impose geometric or prompt-based structure. With strong pretrained features, however, simple fixed-feature heads can be competitive [34]. FSA [21] limits adaptation to the first session, while RanPAC [19] and RanDumb [23] use analytic heads over fixed feature expansions. We freeze the encoder in every session and isolate uncertainty in the resulting head.

Support-only estimation. Cosine NCM [20] and prototypical networks [25] represent a class by its support mean. TEEN [31], Distribution Calibration [36], and higher-order calibration [6] transfer base-class structure to scarce classes; EPT [10] instead learns prototype offsets. FeCAM [5] and HyCal [13] use covariance-aware distances, whereas ULA [33] adjusts base–novel logits. Distribution Calibration synthesizes features, higher-order calibration combines novel and semantically related base covariances after base-session adapter tuning, and EPT optimizes learned offsets. These are relevant calibration mechanisms but have different adaptation or synthetic data budgets from our fixed-feature closed-form head. ShrinkSel does not propose a new covariance estimator. Rather it selects prototype radius jointly with classical covariance shrinkage [12], using class-disjoint base episodes, which retains cosine and shared LDA as exact endpoints.

Different modality budgets. PriViLege [22], FDR [35], BiMC [3], TIM++ [14], and Transductive CLIP [18] use text, prompts, language-derived initialization, or vision–text probability features. They provide important multimodal context but are not information-matched to our frozen image-only heads; the primary rankings therefore keep them separate.

Transductive inference. Transductive few-shot methods use the unlabeled query set to refine class prototypes or

predictions. TIM [2] optimizes mutual information, LaplacianShot [42] enforces graph smoothness, PT-MAP [9] uses optimal transport, and ProtoLP [41] alternates propagation with prototype refinement. These methods are usually studied in balanced, low-way, all-novel episodes and can depend on assumed query proportions [28]. Prototype Rectification [16] also uses unlabeled queries, combining label propagation and feature shifting in episodic few-shot learning. We therefore do not claim the first use of unlabeled queries for prototype rectification. Our QPR targets the asymmetric all-seen FSCIL setting: many-shot base prototypes remain fixed, only gated novel directions move, the support-selected radius and metric are preserved, and no class marginal is provided.

Diagnosis. Approximation-risk analysis [32] studies transfer, margins, and base–novel balance in FSCIL. Our narrower question begins after the representation is fixed. We keep the features fixed, first replace the few-shot novel-class means with means computed from all labeled training samples, and then switch to a linear classifier. The accuracy changes show the contribution of each step in this sequence. Because the two changes can interact, a different order could give a different breakdown of the accuracy gap.

3. Diagnosing and Correcting Prototype Error

3.1. Setting and information regimes

Session 0 provides labeled base classes Y_0; session $t > 0$ adds a disjoint set Y_t with K labeled examples per class. Prediction after session t covers $\mathcal{C}_t = \bigcup_{j=0}^{t} Y_j$. A frozen image encoder produces

$$z(x) = f_{\text{img}}(x)/\|f_{\text{img}}(x)\|_2 \in \mathbb{R}^D. \tag{1}$$

Our CLIP pipeline calls only the image tower. We call Y_t, $t > 0$, *head-novel*: these identities are new to the downstream classifier, but their absence from encoder pretraining cannot generally be verified.

A *support-only* rule may use $z(x)$, labeled supports, and base statistics. A *query-assisted* rule may also use an unlabeled cohort $\mathcal{U}_t$ containing only classes in $\mathcal{C}_t$ and is therefore batch-dependent. Complete labeled novel data are restricted to the retrospective diagnosis. "Training-free" below means no gradient-based parameter fitting; closed-form base statistics and base-only selection remain allowed. Known query-class proportions are not evaluated as a deployable method.

3.2. Ordered matched counterfactual path

For support features S_c^K and the complete training split T_c, define

$$\hat{\mu}_c^K = \text{norm}\left(\sum_{z \in S_c^K} z\right), \qquad \hat{\mu}_c^{\text{full}} = \text{norm}\left(\sum_{z \in T_c} z\right), \tag{2}$$

where $\text{norm}(v) = v/\|v\|_2$ for $v \neq 0$. The complete-split mean is a finite labeled reference, not a population oracle. At the final session, A_n^K and A_n^{full} are class-balanced novel accuracies from cosine NCM with base means fixed and only novel means changed. Let A_n^{LP} be the accuracy of a fixed ℓ_2-regularized multinomial probe trained on the same complete features for all seen classes. The ordered path obeys

$$\underbrace{A_n^{\text{LP}} - A_n^K}_{\text{matched head gap}} = \underbrace{A_n^{\text{full}} - A_n^K}_{\text{mean replacement}} + \underbrace{A_n^{\text{LP}} - A_n^{\text{full}}}_{\text{residual rule change}}. \tag{3}$$

This is an exact accounting identity, not a unique causal decomposition. We report the ordered share $r = (A_n^{\text{full}} - A_n^K)/(A_n^{\text{LP}} - A_n^K)$ only when a crossed support-by-class bootstrap interval excludes a zero denominator. Probe, bootstrap, repair, and classwise definitions are in the supplement.

3.3. Finite-shot motivation

For an unnormalized mean $\hat{\mu}_c = \mu_c + \epsilon$ with $\mathbb{E}[\epsilon] = 0$, $\text{Cov}(\epsilon) = \Sigma/K$, and ϵ independent of the query feature x, let G be a fixed positive-definite metric and $d_G(a, b) = (a - b)^\top G(a - b)$. Then

$$\mathbb{E}[d_G(x, \hat{\mu}_c) - d_G(x, \mu_c)] = \text{tr}(G\Sigma)/K. \tag{4}$$

This states only an expected correct-class distance penalty. It is neither an accuracy bound nor a model of normalized prototypes, competing classes, or estimated G. Because G also changes between-class separation, it motivates selecting the discrimination–noise trade-off rather than prescribing a shrinkage value.

3.4. Support-only geometry selection

ShrinkSel estimates the pooled within-class covariance from complete base features,

$$\begin{aligned} \bar{z}_c^B &= |T_c|^{-1} \sum_{z \in T_c} z, \\ \bar{\Sigma}_B &= \frac{1}{M} \sum_{c \in Y_0} \sum_{z \in T_c} (z - \bar{z}_c^B)(z - \bar{z}_c^B)^\top, \\ \delta &= \max\{\text{tr}(\bar{\Sigma}_B)/D, 10^{-12}\}, \end{aligned} \tag{5}$$

where T_c is the complete base training set and $M = \sum_c (|T_c| - 1)$. This covariance is frozen after the base session.

For n i.i.d. unit support features with sample mean $\bar{z}_c$ and population mean μ_c, the raw resultant length is upward biased because

$$\mathbb{E}\|\bar{z}_c\|_2^2 = \frac{1}{n} + \frac{n-1}{n}\|\mu_c\|_2^2. \tag{6}$$

For $n \geq 2$, we therefore use

$$\hat{r}_c = \sqrt{\left[\frac{n\|\bar{z}_c\|_2^2 - n^{-1}\sum_i \|z_i\|_2^2}{n-1}\right]_+}; \qquad \hat{r}_c = \|\bar{z}_c\|_2 \text{ if } n = 1. \tag{7}$$

Here $[a]_+ = \max(a, 0)$. Clipping ensures a valid radius but can reintroduce finite-sample bias. The prototype family is unit $u_c = \mathrm{norm}(\bar{z}_c)$, raw $\bar{z}_c$, or corrected $\hat{r}_c u_c$. Each is crossed with

$$\begin{aligned} G_\lambda &= [(1-\lambda)\bar{\Sigma}_B + (\lambda+\gamma)\delta I]^{-1}, \quad \gamma = 1, \\ s_c(z) &= -\tfrac{1}{2}(z-p_c)^\top G_\lambda (z-p_c), \\ \lambda &\in \{0, .1, .25, .5, .75, .9, 1\}. \end{aligned} \tag{8}$$

Unit prototypes at $\lambda = 1$ reproduce cosine NCM, while raw prototypes at $\lambda = 0$ reproduce ridge-regularized shared-covariance LDA with equal class priors.

Base-only selection. Let $B = |Y_0|$, let each of T increments contain m classes, and define the class-count weights

$$w_n = \frac{1}{T+1}\sum_{t=0}^{T}\frac{tm}{B+tm}, \qquad w_b = 1 - w_n. \tag{9}$$

For candidate $h = (p, \lambda)$, each pseudo-increment measures anchor and pseudo-novel accuracy, $A_b(h)$ and $A_n(h)$, and scores $J(h) = w_b A_b(h) + w_n A_n(h)$. Base identities are first split into construction, selection, and safety partitions. For ShrinkSel, construction and selection identities are pooled and deterministically repartitioned into four class-disjoint pseudo-novel folds; safety identities never enter selection. Across 200 episodes, a candidate is feasible only if it does not trail raw $\lambda = 0$ LDA in any fold. The feasible candidate with highest pooled J is checked on safety; a regression beyond 0.10 point deploys the exact LDA endpoint. The chosen geometry is refit on authorized base features and frozen. Split sizes, seeds, episode construction, tie-breaking, and the cumulative-count stress test are detailed in the supplement.

3.5. Query-assisted prototype rectification

Query Prototype Rectification (QPR) updates a fixed support-only head. Our primary pipeline uses ShrinkSel; Table 2 also applies the same update to cosine NCM, using unit-norm prototypes and cosine scores. Initial scores define

$$\pi(c \mid z) = \mathrm{softmax}_{c \in \mathcal{C}_t}(\kappa s_c(z)), \quad m_n(z) = \sum_{c \in \mathcal{C}_t \setminus Y_0} \pi(c \mid z). \tag{10}$$

A novel class c admits

$$\begin{aligned} \mathcal{G}_c = \{z \in \mathcal{U}_t :\ & \arg\max_{d \in \mathcal{C}_t} \pi(d \mid z) \notin Y_0, \\ & m_n(z) \geq \tau_q, \\ & c = \arg\max_{d \in \mathcal{C}_t \setminus Y_0} \pi(d \mid z)\}. \end{aligned} \tag{11}$$

If $|\mathcal{G}_c| < m_{\min}$, its prototype is unchanged. Otherwise, define $a_c(z) = \pi(c \mid z) / \sum_{u \in \mathcal{G}_c} \pi(c \mid u)$ and perform

$$\begin{aligned} \tilde{\mu}_c &= \mathrm{norm}\left(\sum_{z \in \mathcal{G}_c} a_c(z) z\right), \\ v_c' &= \mathrm{norm}((1-\alpha)\,\mathrm{norm}(p_c) + \alpha\tilde{\mu}_c), \\ p_c' &= \|p_c\|_2 v_c'. \end{aligned} \tag{12}$$

Base prototypes, non-admitted novel prototypes, and G_λ remain fixed; all queries are then rescored with the same support-only rule: Eq. (8) for ShrinkSel, or cosine scoring for the cosine NCM comparison. The update is one-shot, restarts from the support-only head each session, and assumes neither labels nor class proportions. Ordinary QPR uses the same cohort to construct and evaluate the update. Exact self-exclusion subtracts query z_i from its admitted aggregate when classifying z_i; other cohort queries remain available, so this is still transductive rather than online inference.

The four controls are selected on identity-disjoint base pseudo-query episodes spanning balanced and deterministic-imbalanced cohorts, with held-out combined and anchor safety gates; failure deploys the exact no-update case $\alpha = 0$. The 96-point grid, gate thresholds, calibration episodes, tie-breaking, vectorized self-exclusion, and cumulative-competition calibration are detailed in the supplement. In Table 2, the audited QPR settings are held fixed when moving from ShrinkSel to cosine NCM.

4. Experiments

4.1. Experimental design

Data and encoders. We use BiMC's [3] five-shot class orders and manifests. CUB-200 [30] uses $100 + 10 \times 10$ classes; CIFAR-100 [11] and miniImageNet [29] use $60 + 8 \times 5$. Aircraft [17] and DTD [4] provide additional domain checks. Results average twenty matched support draws, with nested supports for $K \in \{1, 2, 5, 10, 20\}$.

The primary matrix uses frozen OpenAI CLIP [24] ViT-B/16, ViT-B/32, and RN50 image features. MAE-ft ViT-B/16 [8] and supervised ImageNet ResNet-50 [7] provide cross-family checks. A ResNet-18 trained only on benchmark base classes provides a weaker control. Novelty refers to the downstream head, not verified absence from pretraining.

Comparators. Support-only comparisons include NCM [20], shared and diagonal LDA, TEEN [31], ULA [33], FeCAM [5], RanPAC [19], RanDumb [23], and an image-only reduction of HyCal [13]. Query-assisted comparisons include TIM-GD [2], LaplacianShot [42], ProtoLP [41], and uniform-marginal MAP-OT$_u$ [9]. Matched comparisons share features, supports, sessions, and queries. Table 2 evaluates every query-assisted method with both cosine NCM and ShrinkSel as support-only references. The added ShrinkSel variants use its prototypes and a common whitening transform of supports and queries, normalized by a base-only scalar. They retain each method's update, including full LaplacianShot's CL2N preprocessing. Existing base-selected settings are held fixed across heads. The supplement describes the original native-pipeline comparisons; Table 2 extends them with the additional head comparisons.

Metrics and uncertainty. The primary metric is the unweighted average of sample-weighted session accuracies, including session 0. The diagnosis instead uses class-balanced novel accuracy at the final session. Paired method intervals use 10,000 support resamples; diagnostic crossed support–class intervals use 5,000 draws. These intervals quantify the resampled sources of variation under the fixed protocol. The supplement gives further metric and implementation details.

4.2. Locating the recoverable error

In every row of Fig. 2, replacing five-shot novel means recovers more than half of the gap to the complete-data probe. Features, base prototypes, candidate classes, and cosine scoring remain fixed during this replacement. The comparison therefore measures the accuracy change from replacing novel means, rather than from jointly changing the representation and classifier. The same majority pattern holds in the cross-family and Aircraft checks, giving sixteen strong encoder–dataset settings.

The subsequent probe does not always improve accuracy. On CUB, the orange step points left because the probe underperforms complete-data cosine. Additional classifier flexibility is therefore not necessarily beneficial after mean replacement. Equation (3) describes this particular ordering of comparisons.

The supporting diagnostics are consistent with a finite-sample explanation. Accuracy improves with increasing support count, and complete-data means repair more decisions than they damage. Angular error is associated with classwise loss on the standard matrix, but not on Aircraft despite its large aggregate improvement. Whether a direction error changes a label also depends on the nearest competing boundary. The matched replacement is our primary evidence; shot and angle diagnostics help interpret it.

4.3. Support-only recovery

Table 1 compares support-only heads on CUB-200. Shared LDA is the relevant reference because it is already a member of the ShrinkSel family. ShrinkSel is slightly below LDA in sessions 0–1 and improves in the remaining sessions. The session-average gains over LDA are $+0.27$, $+0.06$, and $+0.29$ percentage points on CUB-200, CIFAR-100, and miniImageNet, respectively.

Figure 3 shows the trajectories on all three benchmarks. ShrinkSel and LDA separate as novel classes accumulate on CUB and miniImageNet, while CIFAR remains nearly tied. RanPAC has a strong base-session result but falls behind as novel classes enter. Base-session accuracy alone does not determine the session average.

The ablations indicate that covariance-aware scoring accounts for most of the improvement over cosine. Radius correction has a smaller effect that depends on the representation. A mean's radius contains concentration information, but its squared value is inflated by finite-sample estimation (Eq. (6)). A metric that gives greater weight to a direction also increases the contribution of estimation noise in that direction, consistent with Eq. (4). We therefore select the radius treatment and metric jointly.

Figure 4 shows how performance and foldwise feasibility determine selection. CUB retains several candidates around moderate shrinkage. CIFAR rejects most candidates because at least one fold regresses, while miniImageNet retains corrected-radius choices near the covariance endpoint. Different parameters can yield similar accuracy; the selector need not recover a unique shrinkage coefficient. The supplement reports target-oracle and leave-one-fold-out diagnostics. Novel-order permutations retain the qualitative result but do not vary the base–novel identity split.

The trainable controls in the supplement also depend on example weighting. Sample-weighted optimization favors abundant base examples, whereas inverse-frequency weighting changes the base–novel balance and can reverse the overall ranking. These comparisons do not establish a universally best classifier under scarce novel supervision.

4.4. What an unlabeled cohort adds

Table 2 separates the contribution of the support-only head from the gain supplied by each query-assisted method. With cosine NCM, QPR adds 0.67, 0.37, and 0.54 session-average points on CUB-200, CIFAR-100, and miniImageNet, respectively. With ShrinkSel, its gains are 1.63, 2.03, and 1.21 points. QPR therefore improves both references, with larger gains from the ShrinkSel head in these experiments.

The ranking also depends on the head and dataset. With ShrinkSel, self-excluded QPR leads on CUB, ordinary QPR leads on miniImageNet, and TIM-GD leads on CIFAR at 76.37, compared with QPR's 76.10. With cosine NCM, TIM-GD leads on CIFAR and miniImageNet. These comparisons

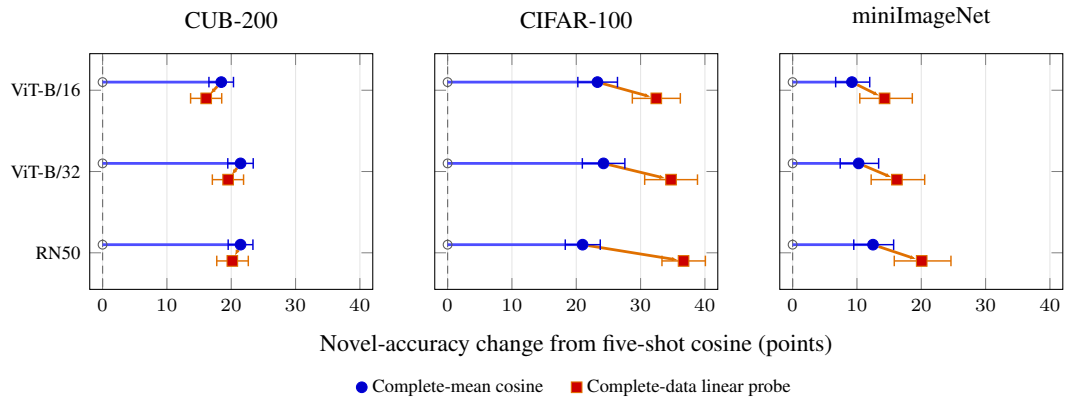

Figure 2. Changes in class-balanced novel accuracy at the final session after replacing novel means and then changing the classifier, across three datasets and three CLIP encoders. Each row starts at zero with five-shot cosine NCM. The blue point replaces only novel means with means from the complete training split. The orange point then switches to a linear probe fitted to the same training images. A leftward arrow means the probe performs worse than complete-mean cosine. Whiskers show 95% two-way bootstrap intervals over support draws and novel classes. The breakdown depends on this comparison order and should not be interpreted as a causal decomposition.

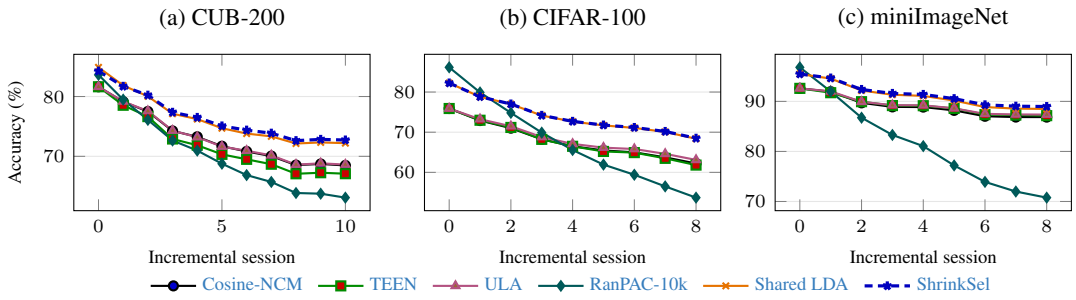

Figure 3. Support-only accuracy as new classes are added on the three standard benchmarks, using frozen CLIP ViT-B/16 features. Each curve averages twenty matched support draws; session 0 is deterministic. Six methods are shown for readability, with the full comparisons in the supplement. Each dataset uses its own vertical scale.

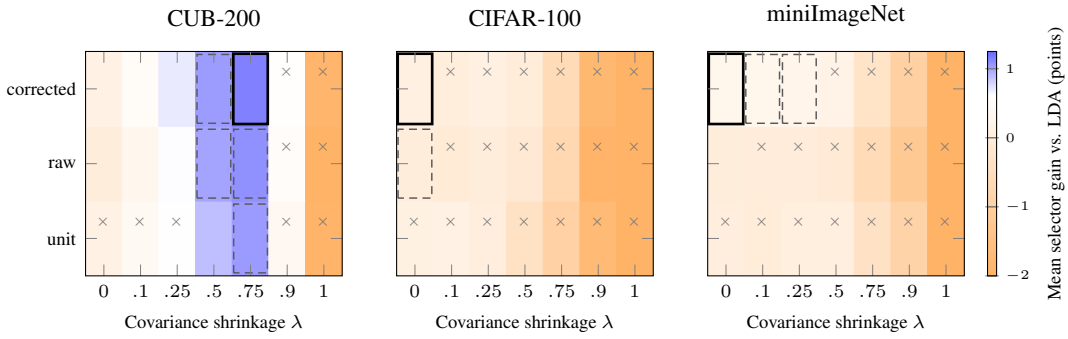

Figure 4. Choosing the ShrinkSel head using CLIP ViT-B/16 base classes. Colors show the mean schedule-weighted gain over shared LDA on class-disjoint base folds; values below -2 points are clipped for readability. A $\times$ marks a candidate that fails the non-regression constraint. The solid box marks the selected candidate, and dashed boxes mark other feasible choices within 0.15 point of its objective.

measure the transfer of fixed base-selected settings across heads; they do not establish a universally best query-assisted update.

Exact self-exclusion retains positive gains on both heads across all three datasets and improves CUB relative to ordinary QPR. A query's inclusion in its own update is therefore

Table 1. Support-only accuracy across CUB-200 sessions with frozen CLIP ViT-B/16 features. Results for sessions 1–10 average twenty matched support draws; session 0 is deterministic. Avg. gives equal weight to each session. PD is the drop from session 0 to the final session and, with a frozen encoder, summarizes changing class competition. Bold and underline mark the best and second-best displayed values. †HyCal denotes the image-only algebraic hard-threshold reduction evaluated here.

Method	Accuracy in each session (%) ↑											Avg. ↑	PD ↓
	0	1	2	3	4	5	6	7	8	9	10		
Cosine-NCM	81.63	79.14	77.49	74.25	73.26	71.69	70.81	70.04	68.53	68.72	68.48	73.09	13.16
TEEN (released)	81.63	78.54	76.64	72.90	71.85	70.33	69.50	68.68	67.08	67.26	67.09	71.95	14.54
ULA (released)	81.63	79.18	77.57	74.30	73.20	71.63	70.93	70.16	68.61	68.83	68.62	73.15	13.01
FeCAM	77.06	71.92	67.35	63.21	60.64	58.29	55.68	53.69	51.68	50.81	49.71	60.00	27.35
HyCal image-only reduction†	79.78	74.06	69.27	64.92	62.10	59.66	56.85	54.87	52.70	51.95	50.80	61.54	28.98
RanDumb (RBF-NCM)	81.56	78.96	77.21	74.02	72.93	71.35	70.47	69.70	68.22	68.35	68.08	72.80	13.49
RanPAC (10k projection)	83.62	79.48	76.09	72.65	70.96	68.77	66.86	65.71	63.86	63.75	63.09	70.44	20.54
Shared-covariance LDA	**84.88**	**81.85**	80.03	77.12	76.23	74.70	73.80	73.34	72.14	72.32	72.20	76.24	12.68
ShrinkSel	84.25	81.71	**80.23**	**77.36**	**76.50**	**75.08**	**74.38**	**73.87**	**72.62**	**72.83**	**72.72**	**76.50**	**11.53**

Table 2. Query-assisted methods compared with cosine NCM and ShrinkSel as support-only references, using frozen CLIP ViT-B/16 features and twenty matched support draws. The support-only row in each block gives the reference accuracy. Avg. is session-average accuracy, and Δ Parent is the paired gain over that reference, with a 95% bootstrap interval. Settings chosen on base classes are kept fixed when changing heads. Bold and underline mark the best and second-best displayed accuracy and gain within each block; displayed ties share formatting. †These variants start from ShrinkSel prototypes in its whitened geometry and retain each method's updates, including full LaplacianShot's CL2N preprocessing. MAP-OT$_u$ assumes a uniform query marginal.

Method	CUB-200		CIFAR-100		miniImageNet	
	Avg. ↑	Δ Parent [95%] ↑	Avg. ↑	Δ Parent [95%] ↑	Avg. ↑	Δ Parent [95%] ↑
Reference head: Cosine-NCM						
Support-only	73.09	—	67.88	—	89.00	—
QPR	73.76	+0.67 [+0.60, +0.73]	68.25	+0.37 [+0.32, +0.42]	89.54	+0.54 [+0.51, +0.58]
QPR (self-excl.)	**73.83**	**+0.73 [+0.68, +0.79]**	68.24	+0.36 [+0.31, +0.41]	89.54	+0.54 [+0.51, +0.58]
TIM-GD	72.99	-0.11 [-0.22, 0.00]	**69.40**	**+1.51 [+1.43, +1.60]**	**90.33**	**+1.33 [+1.28, +1.38]**
LapShot-full	69.71	-3.38 [-3.49, -3.27]	63.18	-4.71 [-4.84, -4.57]	86.02	-2.98 [-3.03, -2.92]
LapShot-core	73.03	-0.06 [-0.07, -0.06]	67.80	-0.08 [-0.09, -0.07]	89.03	+0.03 [+0.02, +0.04]
ProtoLP	66.90	-6.20 [-6.43, -5.97]	62.75	-5.14 [-5.24, -5.03]	86.71	-2.28 [-2.38, -2.18]
MAP-OT$_u$	58.13	-14.96 [-15.11, -14.80]	59.35	-8.53 [-8.64, -8.42]	78.35	-10.65 [-10.84, -10.46]
Reference head: ShrinkSel						
Support-only	76.50	—	74.07	—	91.48	—
QPR	78.13	+1.63 [+1.53, +1.71]	76.10	+2.03 [+1.93, +2.15]	**92.69**	**+1.21 [+1.15, +1.28]**
QPR (self-excl.)	**78.35**	**+1.84 [+1.73, +1.94]**	76.10	+2.03 [+1.91, +2.14]	92.67	+1.19 [+1.12, +1.26]
TIM-GD†	74.78	-1.72 [-1.87, -1.58]	**76.37**	**+2.30 [+2.18, +2.41]**	92.59	+1.11 [+1.03, +1.19]
LapShot-full†	76.66	+0.16 [+0.09, +0.22]	72.87	-1.20 [-1.31, -1.08]	91.37	-0.11 [-0.19, -0.03]
LapShot-core†	76.54	+0.03 [+0.03, +0.04]	74.05	-0.02 [-0.03, -0.02]	91.50	+0.01 [0.00, +0.02]
ProtoLP†	71.56	-4.94 [-5.15, -4.74]	71.81	-2.26 [-2.41, -2.11]	90.61	-0.87 [-0.99, -0.76]
MAP-OT$_u$†	59.03	-17.48 [-17.72, -17.21]	70.84	-3.23 [-3.40, -3.06]	89.67	-1.81 [-1.95, -1.67]

unnecessary for the observed benefit. Other queries remain available, so this is still a transductive evaluation.

The transferred methods are commonly evaluated in balanced low-way episodes. All-seen FSCIL instead combines many-shot base classes with a growing few-shot novel block. The native five-way check confirms that TIM-GD improves in that setting, but does not isolate why its all-seen gains vary across heads and datasets. QPR targets the base–novel asymmetry by updating only eligible novel directions. Those updates can also move a novel boundary across base queries, producing a base-accuracy cost.

In the reported ablations, re-estimating radii from pseudo-

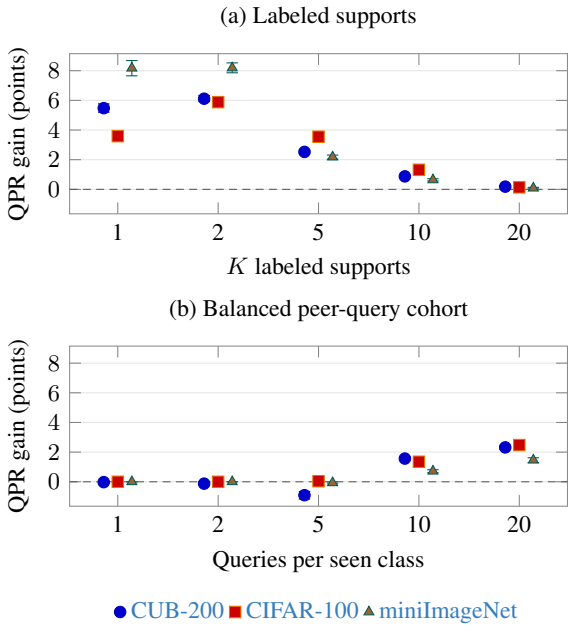

Figure 5. How QPR's gain changes with labeled support and query-cohort size on CLIP ViT-B/16. The top panel uses the complete all-seen query cohort and varies the number of labeled supports. The bottom fixes $K = 5$ and varies the number of queries per class in a balanced cohort. Both show final-session overall (Full) accuracy gains over ShrinkSel, with paired 95% bootstrap intervals over twenty support draws. The same base-selected QPR settings are used throughout.

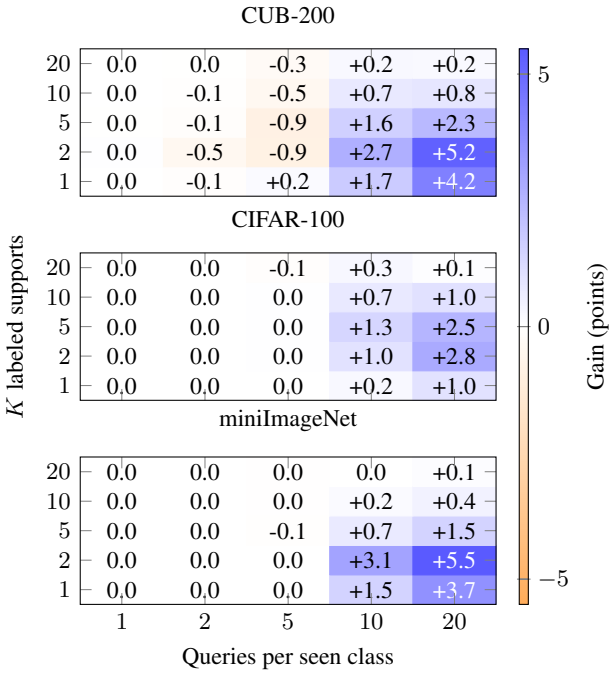

Figure 6. QPR gains for different combinations of labeled support and balanced query-cohort size at the final session. Each cell shows QPR minus ShrinkSel in overall (Full) accuracy, measured in percentage points. Labels are used only to construct the balanced cohort and are hidden from both methods during inference. All 75 conditions use the same QPR settings selected with five labeled supports. This exploratory grid does not test significance separately for each cell.

labeled queries reduces performance on all three datasets. Uniform within-group weights have little effect, whereas removing all conservative controls degrades the update. The results support preserving the selected geometry without establishing that every gate is necessary on every dataset.

Figure 5 shows how support count and cohort size affect the gain. CUB and CIFAR peak at two supports; miniImageNet is similar at one and two. More supports leave less estimation error for QPR to repair. Very small balanced cohorts often yield little or no change, while five-query cohorts can be harmful. Ten or twenty queries per class provide positive gains in the reported cohort sweep.

Figure 6 shows the interaction. Larger cohorts help most when labeled supports are scarce, and their benefit contracts as support count increases. The intermediate five-query region is adverse on CUB. These observations are consistent with a trade-off between remaining prototype error and the reliability of query assignments. The CLIP backbone extensions show the same qualitative pattern. On DTD, QPR improves its parent even though ShrinkSel itself underperforms LDA.

4.5. Boundary and cost checks

The base-only ResNet-18 control does not satisfy the majority criterion on the standard datasets. Both representation quality and pretraining exposure differ from the stronger encoders, so this result does not isolate their effects. Resource measurements use cached ViT-B/16 features. ShrinkSel stores a pooled eigensystem and class prototypes, and adds novel means in closed form. The measurements in the supplement describe the head after feature extraction, not end-to-end image-processing latency. QPR additionally requires the query cohort.

5. Limitations

Our released encoders establish novelty only at the incremental head. CLIP's corpus cannot be audited, and the additional encoder families use ImageNet supervision. The base-only ResNet control excludes novel benchmark classes but is also weaker, so it cannot separate exposure from representation quality.

The complete-data means and linear probe are retrospective references. Their accounting is exact for the chosen

order, but the two terms are descriptive rather than causal. ShrinkSel is a conservative combination of classical estimators; some gains over LDA are practically small, its selector only approximates cumulative novel competition, and transfer is conditioned on a fixed base/novel identity split.

QPR is batch transductive, not an online or support-only method. Other cohort queries remain available even under query-wise self-exclusion, and moving novel prototypes can harm base predictions. Table 2 transfers existing base-selected settings between heads; it does not separately recalibrate each method for each head.

6. Conclusion

Matched counterfactuals across strong frozen representations identify novel-mean replacement as the larger component of the tested ordered head gap. This diagnosis motivates two deliberately bounded corrections: ShrinkSel selects support-only radius and covariance geometry with an exact LDA fallback, while the new QPR method uses an unlabeled cohort to repair novel directions without changing that geometry. Their behavior shows that frozen-feature FSCIL is governed less by a universally superior classifier than by which evidence is available to estimate each scarce class reliably. The two-head comparison shows that QPR improves both support-only references, while the ranking among query-assisted methods depends on the head and dataset.

References

[1] Noor Ahmed, Anna Kukleva, and Bernt Schiele. OrCo: Towards better generalization via orthogonality and contrast for few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 28762–28771, 2024. 2

[2] Malik Boudiaf, Imtiaz Ziko, Jérôme Rony, Jose Dolz, Pablo Piantanida, and Ismail Ben Ayed. Information maximization for few-shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 2445–2457, 2020. 3, 5

[3] Yiyang Chen, Tianyu Ding, Lei Wang, Jing Huo, Yang Gao, and Wenbin Li. Enhancing few-shot class-incremental learning via training-free bi-level modality calibration. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9881–9890, 2025. 2, 4

[4] Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3606–3613, 2014. 4

[5] Dipam Goswami, Yuyang Liu, Bartłomiej Twardowski, and Joost van de Weijer. FeCAM: Exploiting the heterogeneity of class distributions in exemplar-free continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 6582–6595, 2023. 2, 5

[6] Dipam Goswami, Bartłomiej Twardowski, and Joost Van De Weijer. Calibrating higher-order statistics for few-shot class-incremental learning with pre-trained vision transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 4075–4084, 2024. 2

[7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016. 4

[8] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009, 2022. 4

[9] Yuqing Hu, Vincent Gripon, and Stéphane Pateux. Leveraging the feature distribution in transfer-based few-shot learning. In *International Conference on Artificial Neural Networks (ICANN)*, pages 487–499, 2021. 3, 5

[10] Shengqin Jiang, Xiaoran Feng, Yuankai Qi, Haokui Zhang, Renlong Hang, Qingshan Liu, Lina Yao, Quan Z. Sheng, and Ming-Hsuan Yang. Unlocking prototype potential: An efficient tuning framework for few-shot class-incremental learning. *arXiv preprint arXiv:2602.05271*, 2026. 2

[11] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. 4

[12] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. 2

[13] Eunju Lee, MiHyeon Kim, JuneHyoung Kwon, Yoonji Lee, JiHyun Kim, Soojin Jang, and YoungBin Kim. HyCal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 29462–29471, 2026. 2, 5

[14] Yingping Li, Yutong Zou, Yunshi Huang, Changzhe Jiao, Xinlin Wang, Shen Peng, Zhang Guo, and Shuiping Gou. TIM++: Transductive information maximization for few-shot CLIP. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6671–6680, 2026. 2

[15] Chenxi Liu, Zhenyi Wang, Tianyi Xiong, Ruibo Chen, Yihan Wu, Junfeng Guo, and Heng Huang. Few-shot class incremental learning with attention-aware self-adaptive prompt. In *European Conference on Computer Vision (ECCV)*, pages 1–18, 2024. 2

[16] Jinlu Liu, Liang Song, and Yongqiang Qin. Prototype rectification for few-shot learning. In *European Conference on Computer Vision (ECCV)*, pages 741–756, 2020. 3

[17] Subhransu Maji, Juho Kannala, Esa Rahtu, Matthew Blaschko, and Andrea Vedaldi. Fine-grained visual classification of aircraft. Technical Report 1306.5151, arXiv, 2013. 4

[18] Ségolène Martin, Yunshi Huang, Fereshteh Shakeri, Jean-Christophe Pesquet, and Ismail Ben Ayed. Transductive zero-shot and few-shot CLIP. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 28816–28826, 2024. 2

[19] Mark D McDonnell, Dong Gong, Amin Parvaneh, Ehsan Abbasnejad, and Anton van den Hengel. RanPAC: Random projections and pre-trained models for continual learning. In *Ad-*

vances in Neural Information Processing Systems (NeurIPS), pages 12022–12053, 2023. 2, 5

[20] Thomas Mensink, Jakob Verbeek, Florent Perronnin, and Gabriela Csurka. Distance-based image classification: Generalizing to new classes at near-zero cost. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 35 (11):2624–2637, 2013. 2, 5

[21] Aristeidis Panos, Yuriko Kobe, Daniel Olmeda Reino, Rahaf Aljundi, and Richard E Turner. First session adaptation: A strong replay-free baseline for class-incremental learning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 18820–18830, 2023. 2

[22] Keon-Hee Park, Kyungwoo Song, and Gyeong-Moon Park. Pre-trained vision and language transformers are few-shot incremental learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23881–23890, 2024. 2

[23] Ameya Prabhu, Shiven Sinha, Ponnurangam Kumaraguru, Philip H. S. Torr, Ozan Sener, and Puneet K. Dokania. RanDumb: Random representations outperform online continually learned representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 37988–38006, 2024. 2, 5

[24] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021. 4

[25] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. 2

[26] Zeyin Song, Yifan Zhao, Yujun Shi, Peixi Peng, Li Yuan, and Yonghong Tian. Learning with fantasy: Semantic-aware virtual contrastive constraint for few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24183–24192, 2023. 1

[27] Xiaoyu Tao, Xiaopeng Hong, Xinyuan Chang, Songlin Dong, Xing Wei, and Yihong Gong. Few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12183–12192, 2020. 1

[28] Olivier Veilleux, Malik Boudiaf, Pablo Piantanida, and Ismail Ben Ayed. Realistic evaluation of transductive few-shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 9290–9302, 2021. 3

[29] Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Koray Kavukcuoglu, and Daan Wierstra. Matching networks for one shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. 4

[30] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The Caltech-UCSD Birds-200-2011 dataset. Technical Report CNS-TR-2011-001, California Institute of Technology, 2011. 4

[31] Qi-Wei Wang, Da-Wei Zhou, Yi-Kai Zhang, De-Chuan Zhan, and Han-Jia Ye. Few-shot class-incremental learning via training-free prototype calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 15060–15076, 2023. 2, 5

[32] Xuan Wang, Zhong Ji, Xiyao Liu, Yanwei Pang, and Jungong Han. On the approximation risk of few-shot class-incremental learning. In *European Conference on Computer Vision (ECCV)*, pages 162–178, 2024. 3

[33] Sungwon Woo, Dongjun Hwang, Shiwon Kim, Junsuk Choe, and Jongho Nang. Training-free uncertainty-guided logit adjustment for few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, pages 7634–7643, 2026. 2, 5

[34] Tz-Ying Wu, Gurumurthy Swaminathan, Zhizhong Li, Avinash Ravichandran, Nuno Vasconcelos, Rahul Bhotika, and Stefano Soatto. Class-incremental learning with strong pre-trained models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9601–9610, 2022. 2

[35] Zongyao Xue, Meina Kan, Shiguang Shan, and Xilin Chen. Feature decomposition-recomposition in large vision-language model for few-shot class-incremental learning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3153–3162, 2025. 2

[36] Shuo Yang, Lu Liu, and Min Xu. Free lunch for few-shot learning: Distribution calibration. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[37] Yibo Yang, Haobo Yuan, Xiangtai Li, Zhouchen Lin, Philip Torr, and Dacheng Tao. Neural collapse inspired feature-classifier alignment for few-shot class-incremental learning. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2

[38] Chi Zhang, Nan Song, Guosheng Lin, Yun Zheng, Pan Pan, and Yinghui Xu. Few-shot incremental learning with continually evolved classifiers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12455–12464, 2021. 1, 2

[39] Jinghua Zhang, Li Liu, Olli Silvén, Matti Pietikäinen, and Dewen Hu. Few-shot class-incremental learning for classification and object detection: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(4):2924–2945, 2025. 1

[40] Da-Wei Zhou, Fu-Yun Wang, Han-Jia Ye, Liang Ma, Shiliang Pu, and De-Chuan Zhan. Forward compatible few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9046–9056, 2022. 1, 2

[41] Hao Zhu and Piotr Koniusz. Transductive few-shot learning with prototype-based label propagation by iterative graph refinement. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23996–24006, 2023. 3, 5

[42] Imtiaz Masud Ziko, Jose Dolz, Eric Granger, and Ismail Ben Ayed. Laplacian regularized few-shot learning. In *International Conference on Machine Learning (ICML)*, pages 11660–11670, 2020. 3, 5